

Extra Practice Solutions

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1 Histograms

- **Top Left** Unimodal, no outliers, symmetric, would prefer mean and st. dev.
- **Top Right** Unimodal, no outliers, skewed right slightly, prefer 5# summary but an argument could be made it's almost symmetric enough for the mean
- **Bottom Left** Unimodal, no outliers, skewed right, 5# summary
- **Bottom Right** Unimodal or possibly bimodal. It truthfully looks like there is a spike and then the rest is a uniform (flat) distribution. Assuming we go with unimodal it would be skewed right with no outliers. Strongly prefer the 5# summary

2 Boxplots

Fun fact: These boxplots are the same distributions the histograms are drawn from, in the corresponding locations (eg both bottom left graphs are Gamma distributions)

- **Top Left** Roughly symmetrical with two outliers to the left hand side
- **Top Right** No outliers and slightly skewed right
- **Bottom Left** Several outliers and heavily skewed right
- **Bottom Right** Roughly symmetrical with no outliers

3 Scatter Plots

Fun Fact: This is real data from penguins off the coast of Antarctica

- **First One** This is hard to tell. It kind of look like there are two different clumps of data making interpretations hard. If we had to fit a best fit line it would probably be slightly negative and very weak. No outliers seem to jump out at me.

It can be tempting to look at the data points in the extreme top left corner and far right side and call them outliers but that is a common problem people make. They look at the extreme edge and want to see outliers. None of those points truly scream “I’m not near other points and don’t follow the pattern!” at me.

- **Second One** Still weak but this one seems linear (enough). I’d start a line in the bottom left and draw it straight up and towards the right. I may be tempted to say there is a slight curve but I prefer to default towards straight lines when possible.

For outliers you could try to argue there are a few around the edges but if any one of the data points is agreed to be an outlier it feels like we will end up with 20 outliers. Eg if the top right most point is an outlier should the point on the bottom right be an outlier? What about the point slightly up and towards the left from that point? Or the two data points in the top in the center? Bottom two in the center? It doesn’t end which makes me think that our data is just spread out and not that there are outliers.

- **Third One** The two different subpopulations have been made more apparent lol. Let’s just focus on the top, left group for the moment. I’d say it’s moderate or weak, positive, and linear. The bottom right group I’d basically say the same thing (moderate or weak, positive, and linear). The two subpopulations seem to act the same, one is just shifted over from the other.

As for outliers, the point in the dead middle I dislike. I dislike it a lot for among other reason I don’t even know if I’m positive which subpopulation it came from. To me it breaks the pattern and tries to ride in between the two groups. I admit it’s not that far away from other data points but it behaves strangely compared to the patterns we are seeing. Besides that one nothing looks too worrisome for me besides the point at the very top maybe. Again, a lot of this is justifying your thinking.